The Application Note Template

This note is the template for the others. It is a working document: everything the collection uses is demonstrated here once, in place, so a writer starting a new note can copy this file, delete the prose, and keep the furniture. It is also the reference for the design language. When two notes disagree about how to draw a keystroke or where to put a caption, this one settles it.

The material divides in two. Sections 1 to 2.2 are about the mechanics: what to copy, what to rename, how to build, and what the page is made of. Sections 3 to 14 walk the constructs themselves, each shown as the source you write and the result you get.

Read Section 1 first and then skim for the construct you need. Nothing here has to be read end to end.

Numerical results in this document are of three kinds, and this paragraph is itself part of the template. Results quoted from a reference are marked as such. Results computed by hand are given with their working, so the reader can check them. Results the C47 test suite asserts are identified by the case that asserts them. Anything that has to be read off real hardware carries a marker in red. Keep this paragraph, edited to fit, in any note that quotes numbers. It tells the reader what a figure is worth.

Contents

REV	DATE	DESCRIPTION
0.1	2026-07-29	DE. First draft. Consolidates the keypress and screen notation of AN0029, AN0030 and AN0031 into `style/c47appnote.sty`, moves the body to Helvetica, and adds callout blocks, listing styles and the shared keyboard maps.
0.2	2026-08-01	DE. The `keying` block: a keystroke sequence too long for prose, set one action per row (issue #632). Program entry re-set in it as the demonstration. Revision rows now open with the reviser's initials, as this table does.

1 Using This Template

1.1 Copy, rename, gut

Three steps.

1. Copy this directory to `docs/appnotes/sources/AN00NN/` and rename the `.ltx` to match the convention in Section 1.2.
2. Edit the four front-matter lines in the preamble. They are the only place the note's number, title and date appear, and everything else follows from them.
3. Delete every section below and write the note. Keep the revision table, the bibliography, the keyboard appendix if the note needs it, and the author block.

The preamble you are keeping is four lines long:

```
\documentclass[canadian]{article}
\usepackage{c47appnote}

\notenumber{032}
\noteshorttitle{Template}
\notedate{2026-07-29}
\notetitle{The Application Note Template}
```

All four feed the masthead and the running heads, so keep the short title short: it has to sit in the middle of a 7.4 pt header between the number and the date.

There is no byline in the masthead and no field for one. The author is named once, in `authorblock` at the back beside the colophon, which is where a reader looks for attribution and licensing together. A name under the title would only repeat it.

> **NOTE**
> Everything the design language provides comes from one file, `sources/style/c47appnote.sty`, and that file is the single copy. Before AN0032 the same 50 lines of palette and keybox definitions were pasted into AN0029, AN0030 and AN0031, with a comment in each saying to consolidate them once all three had landed. This is that consolidation. Fix a macro once and every note gets the fix.

1.2 The filename convention

Two names, and they do not match each other. The source keeps the long hyphenated form, and the published PDF keeps the older spaced form the collection has always used.

```
source   docs/appnotes/sources/AN00NN/
           AN-00NN-C47-R47-Application-Note-YYYY-MM-DD-XX-short-slug.ltx
PDF      docs/appnotes/
           AN00NN YYYY-MM-DD C47 R47 XX Short Title.pdf
```

`XX` is the author's initials, and getting them wrong credits the work to somebody else. The map:

Table 1 Author initials as they appear in the collection

INITIALS	AUTHOR	NOTES
JM	Jaco Mostert	The bulk of the collection
DL	Didier Lachieze	AN0021 Printing, AN0026 42S Compatibility
DA	David Abram	AN0027 Structural conversions
WGR	William Gordon Rutherdale	AN0023 Elliptic integrals
NUK	Nigel UK	AN0000 Installation
DE	David Emerson	AN0029 First Steps, AN0030 Uncertainty, AN0031 Algebraic Numbers

> **WARNING**
> There are two Davids. `DA` is David Abram and `DE` is David Emerson. AN0030 and AN0031 were both filed as `DA` and had to be corrected. Never copy the initials from a neighbouring note's filename; check who wrote that one first.

A publication with more than one contributor carries every contributor's initials, in the order their work appears in it. AN0023 is published as `AN0023 2026-04-17 C47 R47 WGR JM Elliptic Incl Anx1.pdf` because Annex 1 is Jaco Mostert's and the note it is bound behind is William Gordon Rutherdale's. The source file keeps only the initials of the author of that source, so `AN-0023-...-WGR-elliptic.ltx` is right even though the PDF built from it says `WGR JM`. The two names disagreeing is not a mistake; the PDF is a bundle and the source is one part of it. Credit the same people, saying who wrote what, in `authors` at the back: Section 14.

The date in both names is the date of the revision being published, not the date the note was started. Bump it when you publish, and add a row to the revision table at the same time.

WARNING

Never let the date come from `today`. A note whose masthead is built from the build clock says something different every time anyone compiles it, and the published PDF then disagrees with its own filename. AN0023 was published this way and its source carried a date four days off the PDF's until the note was moved onto this template. `notedate` is the one place the date belongs.

1.3 Building

The style file lives one level up from the note, so `TEXINPUTS` has to point at it. From the note's own directory:

```
TEXINPUTS=../style: lualatex AN-00NN-...-slug.ltx
TEXINPUTS=../style: lualatex AN-00NN-...-slug.ltx
```

Twice, so the table of contents, the `lastpage` count and the cross references settle. The trailing colon on `TEXINPUTS` matters: it is what keeps the default search path.

`lualatex` is required, not optional. The package calls `RequireLuaTeX` and will stop under `pdflatex` or `xelatex`.

---

2 The Page

2.1 Typography

Body and headings are TeX Gyre Heros, which is the metric-compatible Helvetica that ships with TeX Live. Using the clone rather than the system Helvetica is deliberate: every contributor gets the same page, including the ones on Linux and the CI runner, and nothing depends on what happens to be installed.

The page is letter with a 1.25 in margin, which puts the measure near 75 characters. That is the usual readability target, and it also leaves the keyboard maps room to sit at close to their natural width instead of being shrunk to fit.

Mathematics is New Computer Modern Sans Math, also from TeX Live, so a variable in a derivation matches the prose around it instead of falling back to a serif face. Compare the sum in Equation 1 with the sentence carrying it. Fira Math was tried first and rejected: its minus sign is undersized, so $n - 1$ reads as an en dash, and its ν is indistinguishable from a v, which matters in any note that uses ν for degrees of freedom.

One typographic accent does all the labelling: `lsc`, a letterspaced uppercase run. It marks the masthead, a table head, a caption label and a callout title, so structural furniture never needs a colour or a rule of its own to be read as furniture.

The display face is JuliaMono, and it is the one font the package cannot supply. The screen and program boxes reproduce the calculator display and need its glyphs literally: `x⇄y`, `x²`, `Σ+`, `√`, `≠`. Install it from `https://juliamono.netlify.app`. Without it the package falls back to DejaVu Sans Mono and warns during the build, and the boxes will be missing glyphs.

Table 2 The three faces and what each is for

ROLE	FACE	USED FOR
Body	TeX Gyre Heros	Prose, headings, tables, captions, keybox labels
Math	NewCM Sans Math	Everything between dollar signs
Display	JuliaMono	`screen`, `prog`, `cmd`, listings: anything quoting the calculator or the source tree

2.2 The palette

Seven colours, and two of them are spoken for.

Table 3 The palette. `fcol` and `gcol` carry a fixed meaning and are never available for anything else.

NAME		HEX	MEANING
`keyink`		1B1B1B	Body, headings, rules, keyboard: nearly everything
`soft`		6B6A65	Furniture and captions, and anything deliberately quiet
`fcol`		DE6E10	The f shift, and everything reached through it
`gcol`		2C6DB5	The g shift, and everything reached through it
`menufill`		E9E7E2	The fill of a soft-key label
`lcdbg`		F2F3EA	The paper colour of the display
`warnink`		B3341F	The one accent: a warning, and the draft markers

Three rules hold the design together, and they are worth stating plainly because breaking any of them is what makes two notes stop looking like each other.

One. `fcol` orange means ”reached through the f shift” and `gcol` blue means ”reached through the g shift”, everywhere, without exception. Nothing else may use them: not a heading, not a rule, not a running head, not a callout. A reader who has learned that on page 2 must not meet an orange mark on page 9 that means something else. This is why the running heads are `soft` grey rather than the blue AN0029 through AN0031 first used.

Two. A filled panel is the calculator's own screen. `screen` and `prog` carry the `lcdbg` wash because they reproduce the display; tables, listings and callouts are unfilled and bounded by hairlines instead. So a fill on the page always means the same thing, and a quotation of the calculator never looks like a quotation of a file.

Three. Rules carry weight in one direction only. 1 pt closes the masthead and opens the author block, 0.7 pt opens and closes a table, and 0.4 pt is every other hairline: the section rules, the header rule, a capture frame, a listing. No double rules and no vertical rules anywhere.

Note what is missing. There is no hue for a note or a tip, because their titles tell them apart and that is cheaper than teaching the reader two more colours. `warnink` is the only accent, spent on the one callout that is a safety signal and on the draft markers of Section 12. Everything else is keyink, `soft`, or a tint of keyink.

3 Keypress Notation

This is the part of the design language worth most, because it is the part a reader learns once and then reads without effort.

3.1 The kinds of key

Six macros, one per way of reaching a function.

On the C47 there is one physical shift key, pressed once for f and twice for g. On the R47 there are two, a yellow f and a blue g. The notation is the same on both, and a note giving C47 keystrokes need not caveat every line; say once which machine the keystrokes are for and point at the keyboard maps.

Table 4 Keypress macros

SOURCE	RESULT	MEANS
`\key{7}`	7	Press that key
`\fkey{MODE}`	f MODE	f shift, then the key whose orange label is MODE
`\gkey{CLK}`	g CLK	g shift, then the key whose blue label is CLK
`\mkey{DEG}`	DEG	A soft key, named by the label the screen prints above it
`\fmkey{U.P}`	f U.P	f shift, then the soft key carrying U.P on its orange row
`\gmkey{REGR}`	g REGR	g shift, then the soft key carrying REGR on its blue row
`\lkey{f}`	f LONG	A long press

WARNING

A softmenu page is eighteen items: six unshifted, then six f-shifted, then six g-shifted. Before drawing a soft key, check the menu array in `src/c47/softmenus.c` and use `mkey`, `fmkey` or `gmkey` to match. Prose like "`ADV PGMSLV`" reads as unshifted and is wrong: `PGMSLV` is on the f row. Drawing the badges on AN0030 and AN0031 caught two errors that bare command names had hidden.

3.2 Keystroke lines

A keystroke line is set on its own, indented, and wraps at the gaps between keys when it runs long. Two forms. `seq` carries the result the display leaves; `keys` is for a step with nothing to show for it yet.

```
\seq{\key{3}\quad\key{ENTER}\quad\key{4}\quad\key{+}}{X:\ \ 7}
\keys{\fkey{DISP}\quad\mkey{FIX}\quad\key{4}}
```

3 ENTER 4 + → `X:  7`

f DISP FIX 4

Separate keys with `\quad` where the reader should pause and with `\ ` where the keys form one token, as the digits of a number do. `\key{1}\key{1}` is two presses of one number; `\key{1}\quad\key{1}` reads as two separate steps.

Use `screen` for a display line quoted inside a sentence, like `X:  7`, and keep it to one line. Anything taller belongs in a `prog` block.

3.3 When prose stops working: the keying block

The inline forms above are the HP-42S manual's own style, and up to a handful of keys inside a sentence they are the most readable notation there is. They stop working when the sequence is a whole program or a long setup: a paragraph with thirty keycaps in it reads as clutter, however carefully it is written. Past that length, set the sequence one action per row in a `keying` block, the keys on the left and what they accomplish on the right:

```
\begin{keying}
    \step{\key{XEQ}\,\key{.}\,\key{.}}{open a new program}
    \step{\fkey{PRGM}}{enter program-entry mode}
\end{keying}
```

XEQ . .	open a new program
f PRGM	enter program-entry mode

An empty right cell is fine where a step needs no comment. Keep the comments to a phrase: the keys are the thing being read, which is why the column is set small and soft. Section 5 shows a complete program keyed in this way.

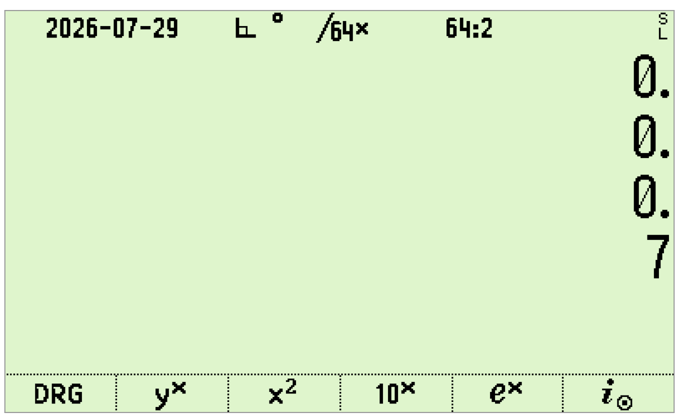
4 A Worked Tutorial

This section is the shape a tutorial should take: state the goal, give the keystrokes, show the screen, say what happened and why. The content is deliberately trivial so the pattern is visible through it.

Add three and four, then set the display to four decimals.

The C47 and R47 use RPN, so there is no equals key. Type the first number, press ENTER to separate it from the second, type the second, then press the operation.

3 ENTER 4 + → `X: 7`

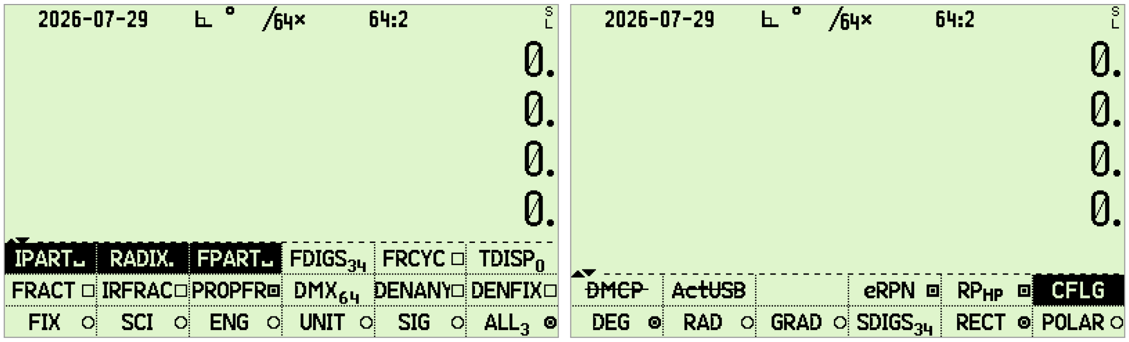


Three plus four. The answer lands in **X**, the bottom line. The tiny **S** over **L** at the top right is the simulator's stack-lift indicator; real hardware carries the battery gauge there instead.

The sum appeared without your saying where to put it. That is the whole of RPN: intermediate results wait on the stack and are consumed by the next operation.

Now the display. Open the display menu with f DISP. The first row offers the six formats, and FIX takes a digit count.

f DISP FIX 4 → `X: 7.0000`



Left, the first page of f DISP: the formats on the unshifted row, the fraction settings on the f row above, and IPART, RADIX and FPART on the g row. Right, the first page of f MODE for comparison. A label with a line struck through it, like DMCP and ActUSB here, is a function the build you are running does not offer.

> **TIP**
> Menu paging wraps. A single press of ↓ from page 1 lands on the last page, which is usually quicker than pressing ↑ twice.

Three things make a tutorial section work. Every keystroke line is complete, so a reader can follow it without inferring a missing ENTER. Every screen shown is a real capture of exactly those keystrokes. And each step says what happened, not just what to press.

---

5 Program Entry

A program is given twice: once as a listing, so the reader can check what they have, and once as the keystrokes that produce it. Both, always. A listing alone leaves the reader guessing where `MVAR` lives; keystrokes alone give them no way to find the step they mistyped.

The listing goes in a `prog` block, in the display face, with the four-digit step numbers the calculator itself shows:

```
0001: LBL 'HYP'
0002:   MVAR 'a'
0003:   MVAR 'b'
0004:   RCL 'a'
0005:   x²
0006:   RCL 'b'
0007:   x²
0008:   +
0009:   √x
0010: END
```

Then the keystrokes, in a `keying` block (Section 3.3), since a program keyed in is exactly the sequence prose cannot carry:

XEQ . .	open a new program
f PRGM	enter program-entry mode
g LBL f α H Y P ENTER	the label
P.FN1 P.FN2 MVAR f α a ENTER	declare the first variable; the menu stays on `P.FN2`
MVAR f α b ENTER	and the second
…	the body, straight from the listing
f PRGM	leave program-entry mode

> **NOTE**
> `MVAR` is not on f ADV, which is where readers look for it. In program-entry mode press P.FN1 on soft key 6, then P.FN2 on soft key 6 again, then MVAR on soft key 4. Say this in full every time; it is the single step readers most often fail to find.

`prog` also takes anything else that wants the display face over several lines: a matrix as the editor shows it, a menu dump, a two-line result. Rows are separated with `\\` and columns lined up with `\` , since the box has no tabbing.

```
y    =  2.1298079
u(y) =  0.0077526
```

---

6 Screen Captures

6.1 The rule

Every capture in a note is a real capture from the simulator, driven by the same keystrokes the note prints, started with `--reset` so it reproduces on any machine. Nothing is mocked up and nothing is retouched. The only processing is a threefold point upscale, which keeps the pixels square in print.

This is not a style preference. A note whose pictures were drawn by hand will drift away from the firmware silently, and the reader has no way to tell. A note whose pictures are captures can be regenerated and diffed.

6.2 Placing one

Three macros, by how the picture is used.

Table 5 Capture macros

SOURCE	USE
`\shot[w]{file}{caption}`	One capture in the flow of the text, centred, not floating, so it stays where the prose put it. Width defaults to 0.58 of the text block.
`\shotpair{l}{r}{caption}`	Two side by side. Each lands at half the measure, so a pair is for comparing the shape of two screens, not for reading labels off them.
`\shotfig[w]{f}{cap}{label}`	A numbered float, for a capture the note cross-refers to from elsewhere.

All three prepend `screens/` to the filename and supply the extension, so the argument is bare: `02-result`, not the path and not the `.png`.

Prefer `shot`. A capture belongs beside the keystrokes that produced it, and a float will wander. Reach for `shotfig` only when the note genuinely cross-refers to the picture, as Figure 1 is referred to from Section 13.

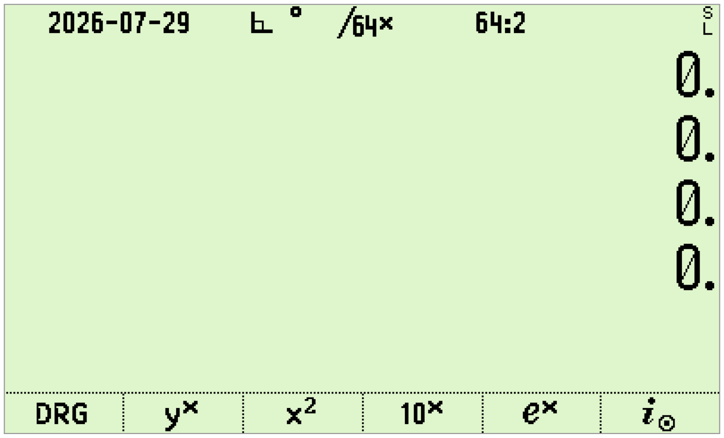


Figure 1 A new calculator, straight out of a reset: an empty stack and MyMenu along the bottom. This is what every capture recipe in `screens/scripts` starts from.

Captions are sentences, in `soft` grey, and they say what to look at. ”The DISP menu” is not a caption. ”The formats on the unshifted row, the fraction settings on the f row above” is.

6.3 Reproducing them

Record the recipe. Every note that carries captures carries a `screens/scripts/README.md` giving the build used, the exact command per picture, and the upscale step. The pattern:

```
./c47 --reset --exec 'press "@k 30"; press "@k 12"; press "@k 23";
                      press "@k 36"; snap s2add'
magick s2add.bmp -filter point -resize 300% screens/02-result.png
```

`press "@k NN"` presses physical key NN, numbered row-major from the C47 table in `src/c47/assign.c`. `press @f` and `press @g` set a pending shift, and `press F1` to `press F6` press the soft keys. A longer sequence goes in a `.t47` script beside the README rather than on the command line.

> **WARNING**
> A plot needs `--snapskiprefresh`, or `snap` redraws the screen and wipes the drawn pixels. The simulator also shows a stack-lift indicator at the top right where hardware shows the battery gauge, so say so in the caption the first time a capture appears in a note.

To check a capture set still reproduces after the firmware has moved, rerun the recipe and diff:

```
compare -metric AE /tmp/regenerated.png screens/02-result.png null:
```

Zero differing pixels is the answer you want.

7 Callout Blocks

Three kinds, and three is the whole set. A page carrying six kinds of coloured box carries none, because the reader has stopped distinguishing them.

> **NOTE**
> An aside the reader can skip without losing the thread. Background, a cross reference, a reason something is the way it is.

> **TIP**
> A shorter route to the same place. A shortcut, a default worth changing, a keystroke that saves three.

> **WARNING**
> A trap that will cost the reader an hour. Something that looks right and is not, a step that silently does nothing, or behaviour that differs between the simulator, the DM42, the DM42n and the R47. Spend these sparingly; a note with nine warnings has none.

They are unfilled, bounded on the left only, and titled in a letterspaced label. A rule and a label are enough to set a paragraph apart, and they leave the page quieter than a shaded panel does. A note and a tip carry the same black rule and are told apart by the title alone; only the warning spends the accent, because only the warning is a safety signal.

They break across pages, so a long one is safe. Keep them to a paragraph or two even so: a callout the size of a section should be a section.

The rule for choosing is what the reader loses by skipping it. Nothing, and it is a note. Time, and it is a tip. An hour, and it is a warning.

There is no separate block for hardware differences, and that is deliberate. A feature the reader's build does not carry, a memory ceiling, a capture that will not reproduce: each of those costs an hour, which is the definition of a warning. Name the build in the first sentence and the reader knows whether the box is about them.

---

8 Code and Text Listings

Two listing styles, both unfilled and closed top and bottom by the same 0.4 pt hairline a capture frame uses. A listing reads as a tabulated block rather than as a shaded panel, which is what keeps it distinct from the `prog` and `screen` boxes: those are washed with `lcdbg` because they reproduce the display, and nothing else on the page is.

`c47text` is the default and takes anything plain: a state-file excerpt, a `.t47` script, a shell command, a menu dump.

```
EQUATIONS
1
p×V=n×R×T
OTHER_CONFIGURATION_STUFF
```

`mathescape` is on, so × inside a listing renders as the multiplication sign the calculator uses. That is how to quote a formula containing a glyph the keyboard has but ASCII does not.

c47code adds C syntax highlighting and line numbers, for firmware source:

```c
/* The relative step assumes the model varies on the scale of x. */
static real_t centralDifference(real_t (*f)(real_t), real_t x) {
  const real_t h = fmax(fabs(x), 1.0) * CBRT_EPSILON;
  return (f(x + h) - f(x - h)) / (2.0 * h);
}
```

Quote source sparingly and quote it exactly. A note is not the place to paraphrase the firmware, and a listing that has drifted from the tree is worse than no listing. Give the path with `file`, as in `src/mathematics/uncertainty.c`, so a reader can go and look.

For one inline fragment use `lit`, which is `\lit` followed by the text between two matching delimiters. It is the right tool for a macro name, a flag, or a line of a file format.

9 Tables

Tables are booktabs throughout: `\toprule` at the top, one `\midrule` under the head, `\bottomrule` at the bottom. Never a vertical rule and never a double rule, because both of them draw the eye to the grid instead of to the numbers. `@{}` on the outer columns is what makes the rules line up with the text block. Small or footnote size, since a table in body size beside Helvetica prose looks loud. Caption above, labelled TABLE and numbered.

Head cells go in `\hd`, which letterspaces them, so the head reads as furniture without needing a rule or a weight change of its own. To group rows, use `\addlinespace`, never another rule: the palette in Table 3 is grouped that way.

`ntable` wraps the whole pattern up, and it is the shortest way to a correct table:

```latex
\begin{ntable}{@{}l l@{}}{The caption sentence.}{tab:label}
    \hd{Source} & \hd{Means} \\ \midrule
    \lit|\key{7}| & Press that key \\
\end{ntable}
```

Table 6 The shape of a house table. Three rules, no verticals, head letterspaced.

COLUMN	TYPE	NOTES
First	`l`	Left, natural width
Second	`l`	Left, natural width
Third	`p{0.42\textwidth}`	Fixed width, so prose wraps rather than pushing the table past the margin

Give every table a caption and a label and refer to it in the prose. A table nobody points at is decoration. When a table runs wide, cut `tabcolsep` to 3 pt and drop to `footnotesize` before you consider landscape.

Command reference tables are their own case, and AN0030 is the model: one small table per command, with rows for what it takes from the stack, what it leaves, what it reads, what it writes, and what it refuses. A reader looking up a command wants the same five rows every time, in the same order.

10 Charts

Charts are `pgfplots`, wrapped in `nchart` so they float and caption like any other figure. The chrome is recessive by default: axis lines on the left and bottom only, no box, horizontal grid only, ticks outside, labels in `soft`. What is left is the data.

10.1 Why a chart here has almost no colour

This is worth the space, because the conclusion is not a preference. A palette that encodes identity has to clear four things at once: a chroma floor, so a hue does not read as grey; a lightness band; a contrast floor against the paper; and a separation floor, measured pairwise, of $\Delta E \geq 15$ in normal vision and ≥ 8 under simulated protanopia and deuteranopia.

Rule 1 of Section 2.2 has already spent orange on the f shift and blue on the g shift, and `warnink` has spent red. Testing what is left against those three reserved inks:

Table 7 Every hue that could have been a second series, and what stopped it. Measured pairwise against `warnink`, `fcol` and `gcol`.

CANDIDATE	VERDICT	WHY
violet `8A5CC7`	unusable	Collapses against `gcol`, ΔE 3.4 under protanopia
ochre `8A6B00`	unusable	Collapses against `warnink`, ΔE 2.5 under deuteranopia
teal `00918A`	unusable	Chroma 0.089, below the 0.10 floor: reads as grey
teal `00A099`	passes	Clears `warnink`, `fcol` and `gcol` on every pair, in normal vision and under all three simulated deficiencies

Exactly one hue survives. It is `chartink`, and a chart gets at most that one, for the single series that is the whole point of the chart. Everything else is `keyink` and the grey ramp.

> **NOTE**
> That is not a hardship. A note that will be printed, photocopied and read on a grey electronic-paper screen wants its series told apart by dash pattern and marker anyway, which is also the only encoding that survives a reader with full colour blindness. The constraint and the right answer coincide.

Magnitude uses the grey ramp `seq1` to `seq5`, which passes the ordinal checks: monotone lightness, an adjacent step of at least $\Delta L = 0.06$, and a light end still clearing 2:1 against white. `seq1` is `keyink` and `seq3` is `soft`, so the ramp is built out of the palette rather than beside it.

the magnitude ramp `chartink`

10.2 Emphasis: one bar is the point

The commonest chart in this collection ranks contributions and says which one to attack. That is not a categorical chart; it is an *emphasis* chart. One mark in full `keyink`, the rest in `seq4`, and the reader's eye goes to the right place without a legend, a hue or a callout.

Sort by magnitude, always, unless the category order carries meaning of its own. An unsorted ranking makes the reader do the sorting.

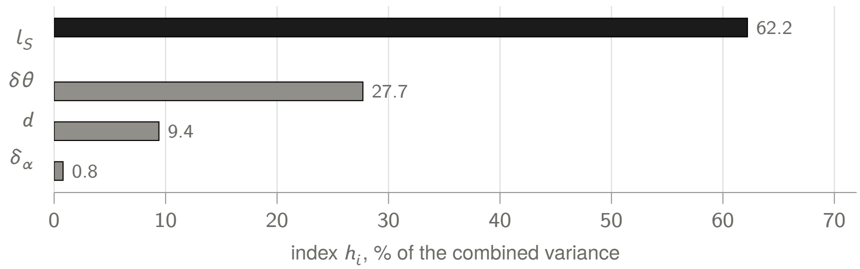


Figure 2 The GUM Annex H.1 uncertainty budget, ranked. l_S carries 62 % of the variance, so it is the only term worth attacking; everything below 10 % is noise. Emphasis, not colour, is what says so.

10.3 One series: the plain case

Most appnote charts have one series, and a single series needs no legend, because the caption names it. Label the axes, give the units, and stop.

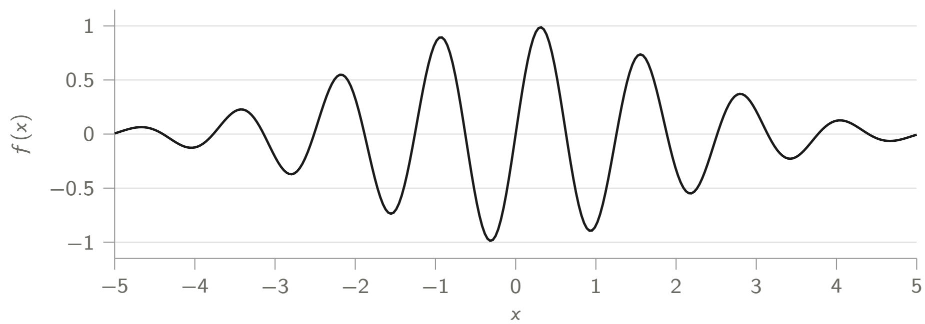


Figure 3 The wave packet $f(x) = \sin(5x)\, e^{-x^2/8}$ of Application Note 29 [3], over $-5 \leq x \leq 5$. One series, so no legend: the caption names it.

10.4 Two or three series: dash, then marker

Identity runs A, B, C in a fixed order and never cycles: `canA` solid, `canB` dashed, `canC` dotted, all in `keyink`. A legend appears as soon as there is more than one series. A fourth series is not a fourth dash pattern; it is a second chart, or a small multiple.

Measured data and a fitted curve are a special case of two series that needs no dash ladder, because points and a line are already distinct: use `canpoints` for the data and `canA` for the fit.

10.5 The rules, in short

- **Never a second y axis.** Two measures of different scale are two charts, or one chart with both series indexed to a common base. This is the single commonest way to mislead a reader with a chart that is technically accurate.
- **Identity by dash and marker, in the fixed order A, B, C.** At most one hue, `chartink`, and only for the series that is the point.
- **Magnitude by the grey ramp**, light to dark, more is darker. Never a rainbow.
- **Emphasis before category.** If the story is ”this one dominates”, that is one dark mark among grey ones, not five hues.
- **Sort by magnitude** unless the category order means something.

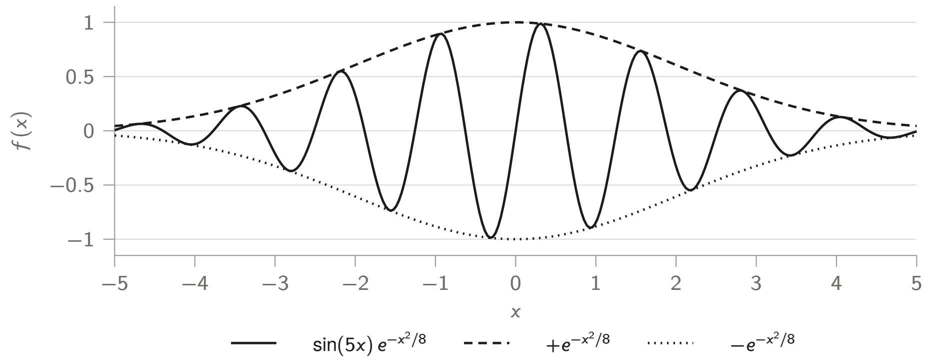


Figure 4 The same packet against the envelope that bounds it, $\pm e^{-x^2/8}$. Solid, dashed and dotted in that fixed order, one ink, and a legend because there is more than one series.

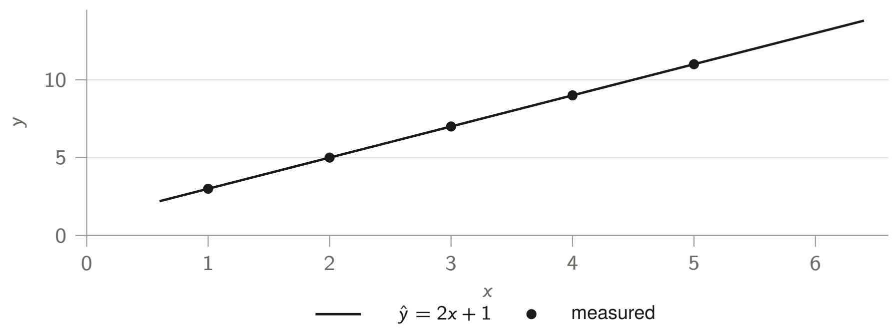


Figure 5 The five pairs of Application Note 29 [3] and the line the regression menu fits to them, $y = 2x + 1$, with $r = 1$. Points against a line need no dash ladder.

- **Label the axes and give the units.** A number without a unit is not a result.
- **No legend for one series**; a legend for two or more, and direct labels where they fit. Never a value on every point.
- **Recessive chrome.** Horizontal grid only, no box, no fill under a line unless the area is the quantity.

---

11 Mathematics

Inline mathematics sits in the sentence: the standard uncertainty of an input estimate x_i is written $u(x_i)$, with $\nu_i = n - 1$ degrees of freedom. Display mathematics that is not referred to again goes unnumbered,

$$u(\bar{q}) = \frac{s(q)}{\sqrt{n}}$$

and anything the prose comes back to gets a number and a label:

$$u_c^2(y) = \sum_{i=1}^{N} c_i^2\, u^2(x_i) + 2 \sum_{i=1}^{N-1} \sum_{j=i+1}^{N} c_i\, c_j\, u(x_i, x_j) \tag{1}$$

Equation 1 is the law of propagation of uncertainty, and the reference is how the reader gets back to it twelve pages later. Number what you cite and nothing else.

Where a derivation matters, give it. Where only the result matters, give the result and cite the source. A note that reproduces four pages of a standard the reader can download has spent four pages badly.

$$s(q) = \sqrt{\frac{1}{n-1}\sum_{k=1}^{n}(q_k - \bar{q})^2} \tag{2}$$

$$u_c(y) = \sqrt{\sum_i c_i^2 u^2(x_i)} \tag{3}$$

Units are set upright and separated from the number: 31.71 nm, 50.000838 mm, 0.029 °C. Percent takes a thin space, 95 %, which is the convention the GUM uses and the collection follows.

---

12 Semantic Markup

Mark up what a thing *is*, not how it should look. Then a later decision about how commands should be set is one edit in the style file instead of a hundred in the notes.

Table 8 Semantic macros

SOURCE	RESULT	FOR
`\cmd{U.CALC}`	`U.CALC`	A command as the catalog names it
`\vr{U.TAB}`	`U.TAB`	A named variable
`\reg{X}`	**X**	A stack register or a labelled display field
`\menu{UNCT}`	`UNCT`	A menu
`\file{src/c47}`	`src/c47`	A path in the source tree
`\swatch{fcol}`		A palette chip, outlined so a pale fill still reads
`\lsc{note}`	NOTE	A letterspaced label: the one typographic accent
`\HW{timing}`	**[HARDWARE: timing]**	A figure still to be read off real hardware
`\TODO{H.2}`	**[TODO: H.2]**	A hole in the draft

`HW` and `TODO` print red on purpose. A draft circulated for comment should make its holes impossible to miss, and a note must not reach master with either still in it. Grep for both before you open the merge request.

---

13 Cross References and Citations

Label anything you point at and point at it by number, never by position. ”Section 7” and ”Figure 1” survive an edit; ”the section above” and ”the figure on page 4” do not.

The label prefixes the collection uses: bare words for sections, as in `\label{callouts}`, and `fig:`, `tab:` and `eq:` for the three numbered kinds.

Cite the other notes. A note that stands alone duplicates one that already exists, and the collection is worth more as a set of cross references than as thirty separate documents. Cite by number in the prose, as in Application Note 29 [3], and give the full title in the bibliography. External standards get a full entry with a URL where one exists [6].

> **TIP**
> Before writing a new note, read the ones next to it. AN0029 was nearly filed as redundant with AN0028 until the question was settled: AN0028 tours the settings for a reader who already has opinions about them, and carries no keystrokes or captures at all, while AN0029 starts at the first switch-on. They cite each other now. That conversation belongs in the note's opening paragraphs.

---

14 Back Matter

In order: any appendices, the keyboard maps if the note gives keystrokes for both machines, the bibliography, then the author block.

`keyboardappendix` emits both maps, each on its own page, labelled `fig:ckbd` and `fig:rkbd` so the body can refer to them. Ninety lines of TikZ used to be pasted into each note that wanted them; now it is one command. Include it in a note that gives keystrokes, and leave it out of one that is purely about the firmware. `ckeyboard` and `rkeyboard` place a single map if you want to lay out the page yourself.

`colophon` takes the firmware version the note was written against and emits a closing page recording what built the document: the three faces and where they come from, the toolchain, the style-file version, and how the captures were made. A note that promises its captures reproduce has to say what they reproduce against, and this is where that promise is recorded.

The author block is `\authorblock{name}{site}{email}` and carries the GFDL notice with it. Every note in the collection is under the GNU Free Documentation License, and both the SPDX line at the top of the source and this block at the bottom are required.

The site and the email are each optional. Leave one or both empty and the block prints what it was given, so an author who has published neither is named and nothing more, which is how AN0023 credits its author. Never fill either field in on somebody's behalf: a contact detail in a published document is theirs to offer.

A note with more than one contributor uses `authors` instead, one `contributor` per line, in the order the work appears in the document. The fourth argument is the part that is theirs, and it is not optional: attribution that does not say who wrote what leaves every reader guessing, and it is the one thing a reader cannot work out for themselves.

```
\authors{
    \contributor{William Gordon Rutherdale}{}{}{This note}
    \contributor{Jaco Mostert}{}{}{Annex 1, the worked tutorial}
}
```

Reach for it whenever the published PDF has more than one hand in it, including the case where the `.ltx` is one author's and an annex bound behind it is another's. The initials in the PDF filename and the lines in this block have to name the same people: Section 1.2.

> **NOTE**
> The first two lines of every source file are the SPDX identifier and the copyright line, then a comment saying what the note is and how to build it. Copy them from this file; they are what the licence audit reads.

---

15 Summary

- Copy this directory, rename the `.ltx` per Section 1.2, edit the four front-matter lines, delete the prose.
- Check the author initials against Table 1. There are two Davids.
- Build twice with `TEXINPUTS=../style: lualatex`, under lualatex, with JuliaMono installed.
- `fcol` orange means the f shift and `gcol` blue means the g shift. Nothing else may use them.
- Check `src/c47/softmenus.c` before drawing a soft key, and use `mkey`, `fmkey` or `gmkey` to match the row. Eighteen items to a page, six per row.
- Every capture is a real `--reset` capture of the keystrokes printed beside it, with the recipe recorded in `screens/scripts/README.md`.
- Give a program as a listing and as keystrokes. Both.
- Three callout kinds, chosen by what the reader loses in skipping the box. A build difference is a warning, named by build in the first sentence.
- Charts tell series apart by dash and marker, in the fixed order A, B, C. One hue is available, `chartink`, and exactly one: Table 7 shows what stopped the others. Never a second y axis.
- Label and cite everything you point at, including the other notes.
- Grep for `TODO` and `HW` before opening the merge request.

The C47 keyboard

Primary function on the key; orange = f (shift once) above; blue = g (shift twice) below.

→I	y^x	x^2	10^x	e^x	α
Σ+	**1/x**	**$\sqrt{x}$**	**LOG**	**LN**	**XEQ**
a b/c	#	.ms	.d	LBL	GTO
|x|	%	π	ASIN	ACOS	ATAN
STO	**RCL**	**R↓**	**SIN**	**COS**	**TAN**
∡	Δ%	$\sqrt[x]{y}$	*i*	→R	→P
COMPLEX		LASTx	MODE	DISP	UNDO
ENTER		**x⇄y**	**+/−**	**EEX**	**←**
CPX		STK	TRG	EXP	CLR

BST	EQN	ADV	MATX	STAT
↑	**7**	**8**	**9**	**÷**
REGS	HOME	FIN	X.FN	PLOT
SST	BASE	CONV	FLAG	PROB
↓	**4**	**5**	**6**	**×**
FLGS	BITS	CLK	REAL	INTS
	ASN	USER	P.FN	PRINT
f / g	**1**	**2**	**3**	**−**
	KEYS	α.FN	LOOP	I/O
OFF	VIEW	SHOW	PRGM	CAT
EXIT	**0**	**·**	**R/S**	**+**
SNAP	STOPW	INFO	TEST	CNST

The R47 keyboard

The R47 has two physical shift keys: a yellow f and a blue g. Primary on the key; f above, g below.

i	$i_{\odot}$	$x!$	$\sqrt[x]{y}$	10^x	e^x
x^2	$\sqrt{x}$	$1/x$	y^x	LOG	LN
→R	→P	.ms	.d	→I	#
$\lvert x \rvert$	%	π	USER		
STO	RCL	R↓	DRG	f	g
∡	Δ%	R↑	ASN		

COMPLEX	LASTx	DISP	PFX	UNDO
ENTER	x⇄y	+/−	EEX	←
CPX	STK	TRG	EXP	CLR
α	SIN	COS	TAN	STAT
XEQ	7	8	9	÷
GTO	ASIN	ACOS	ATAN	PLOT
BST	BASE	INTS	MATX	EQN
↑	4	5	6	×
REGS	BITS	REAL	X.FN	ADV
SST	PREF	CONV	FLAG	PROB
↓	1	2	3	−
FLGS	KEYS	CLK	α.FN	FIN
OFF	VIEW	SHOW	PRGM	CAT
EXIT	0	·	R/S	+
INFO	I/O	a b/c	P.FN	CNST

References

[1] C47/R47 Application Note 018, *Keyboard*.

[2] C47/R47 Application Note 028, *Configurability Tour*.

[3] C47/R47 Application Note 029, *First Steps*.

[4] C47/R47 Application Note 030, *Measurement Uncertainty*.

[5] C47/R47 Application Note 031, *Algebraic Number Identification*.

[6] JCGM 100:2008, *Evaluation of measurement data – Guide to the expression of uncertainty in measurement*, Joint Committee for Guides in Metrology, 2008. `https://www.bipm.org/en/committees/jc/jcgm/publications`

[7] C47/R47 function reference. `https://47calc.com/documentation`

[8] C47 developer wiki. `https://gitlab.com/rpncalculators/c43/-/wikis/home`

[9] C47 community wiki. `https://gitlab.com/h2x/c47-wiki`

Colophon

Body and headings	TeX Gyre Heros, the metric-compatible Helvetica from TeX Live
Mathematics	New Computer Modern Sans Math, from TeX Live
Display face	JuliaMono, from juliamono.netlify.app
Typeset with	`lualatex`, TeX Live 2025
Design language	`c47appnote` v1.1, `docs/appnotes/sources/style`
Charts	`pgfplots`, one hue and the grey magnitude ramp
Screen captures	the GTK simulator at `--reset`, upscaled 3× with a point filter
Written against	firmware `0.109.03.04`

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